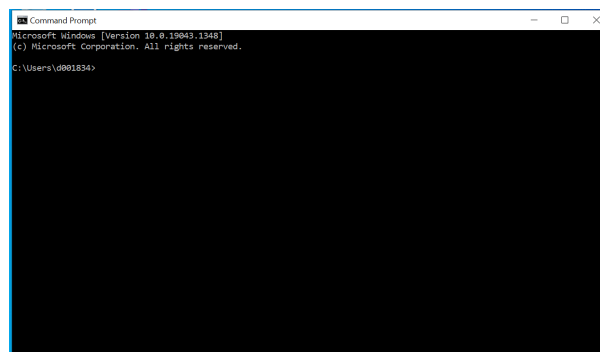


Short guide for Python installation on Windows

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1. Download the executable file from <https://www.python.org/downloads/> (current release Python 3.12.6). This file is typically saved in the Windows Download folder.
2. Execute the downloaded file: in the window that appears, first select **Add python to path** (at the bottom), then do the installation (at the top).
3. The software is installed in folder
`C:\Users\yourname\AppData\Local\Programs\Python\Python312`
4. Open the Windows Command prompt



5. Using command `cd` move to folder
`C:\Users\yourname\AppData\Local\Programs\Python\Python312\Scripts`
where you find file `pip3` (or use the file manager to move to the folder, then, with the right click, open the terminal/command window).
6. Execute the following command line: `pip3 install virtualenv` which creates file `virtualenv` in folder
`C:\Users\yourname\AppData\Local\Programs\Python\Python312\Scripts`
7. Now create the virtual environment. Move to `C:\Users\yourname`. From the Windows command prompt, being in folder `C:\Users\yourname`, execute the command:

```
$C:\Users\yourname\AppData\Local\Programs\Python\Python312\python -m venv pyvenv1
```

which generates folder

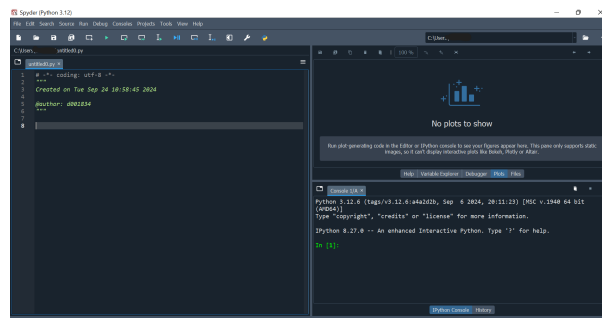
```
C:\Users\yourname\pyvenv1
```

8. Using `cd` in the Windows Command prompt, move to folder
`C:\Users\yourname\pyvenv1\Scripts`
and execute the command: `C:\Users\yourname\pyvenv1\Scripts>activate.bat`
(you activate the virtual environment stored in `pyvenv1`)

9. Now you see in the command window
(pyvenv1) C:\Users\yourname\pyvenv1\Scripts>
which means that the virtual environment is active.
10. Now use `pip3` to install the packages we need (in the order you see below, and all of them):
 - (a) (pyvenv1) C:\Users\yourname\pyvenv1\Scripts> `pip3 install numpy`
 - (b) (pyvenv1) C:\Users\yourname\pyvenv1\Scripts> `pip3 install matplotlib`
 - (c) (pyvenv1) C:\Users\yourname\pyvenv1\Scripts> `pip3 install pandas`
 - (d) (pyvenv1) C:\Users\yourname\pyvenv1\Scripts> `pip3 install scikit-learn`
 - (e) (pyvenv1) C:\Users\yourname\pyvenv1\Scripts> `pip3 install spyder`

In folder `pyvenv1\Lib\site-packages` you'll find the packages you just installed (plus all the dependencies).

11. In the Command prompt window execute the command:
`C:\Users\yourname\pyvenv1\Scripts>deactivate` (you deactivate the virtual environment).
12. Spyder is in folder `C:\Users\yourname\pyvenv1\Scripts`. Use the file manager to find file `spyder.exe`, select it with the mouse, right click and select **pin to taskbar**. You'll see the Spyder icon on the bottom of your screen. Click on this icon (and wait.... it might take one or two minutes, it depends on the computer). You'll see the following window:



which is the IDE for Python.

If you need another virtual environment for another class you can start again from step 7 (of course use another name for the virtual environment, like `pyvenv2`).